ORGANIC INTELLIGENCE PROTOCOL (OIP)

Consolidated Master Technical Reference

Archival Evidence Record and Governance Specification

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Governing principle

No evidence, no operationalization.

This document consolidates the recorded OIP methodology, version lineage, audit states, mathematical notation, and claim boundaries. It distinguishes directly checked artifacts from reported results and unresolved evidence. It does not certify OIP as empirically valid, production-ready, or independently security-validated.

Public research records:

https://zenodo.org/records/23635920

https://github.com/sudharsandas27-ux/Sudharsan-Das

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1 Purpose, Scope, and Scientific Status

The Organic Intelligence Protocol (OIP) is a research-stage methodology for governing evidence qualification, construct measurement, dataset qualification, analytical authorization, and the limits of downstream claims. This edition consolidates the documented conceptual lineage from OIP v1.0.26 through v1.0.39 and records the separate synthetic software reference implementation OIP v0.2.0.

This document is a technical reference and archival synthesis. It is not a new dataset audit, a proof of effectiveness, a security certification, or a grant of commercial rights. Historical results are not silently promoted into independently verified findings when the underlying executed notebook, dataset, manifest, or source artifact is unavailable for reconstruction.

1.1 Evidence labels used in this document

Verified finding A statement directly checked against an available source artifact or recalculated from an available file in the preparation of this edition. Verification is limited to the artifact and operation stated.

Reported result A result described in an archived report or summary but not independently reconstructed from its underlying runtime, raw data, or complete primary source package for this edition.

Unresolved evidence
A conflict, missing source, unverified count, or blocked qualification state for which the available material does not authorize a definitive reconciliation.

Claim not established
A scientific, operational, commercial, or security assertion that the supplied evidence does not establish.

A SHA-256 match establishes byte-level identity relative to the compared file. It does not establish scientific correctness, source authenticity, immutability of storage, or validity of the method.

1.2 Scope exclusions

This edition does not establish empirical, predictive, causal, universal, or commercial validity; superiority to alternative methods; production readiness; hardware authorization; or verified protection against real-world agent failures. Those claims require separate, appropriately designed evidence.

2 Core Philosophy and Governing Sequence

$$\boxed{\text{Evidence} \longrightarrow \text{Measurement} \longrightarrow \text{Score} \longrightarrow \text{Evaluation} \longrightarrow \text{Claim}}$$

Evidence must authorize measurement. Measurement must authorize scoring. Scoring must precede evaluation. Evaluation must precede a scientific claim.

> **No evidence, no operationalization.**

When required evidence is missing, contradictory, insufficient, or unverified, the affected downstream operation remains blocked or pending. The protocol does not authorize unsupported proxy substitution, silent imputation, artificial outcome construction, or post-hoc tuning to manufacture a preferred result.

2.1 Canonical evidence classes

1. **Verified Findings.** Findings directly supported by the specific source artifact or computation identified in the record.
2. **Reported Results.** Findings stated in earlier audit records where full independent reconstruction has not been performed for this edition.
3. **Unresolved Evidence.** Conflicts and qualification gaps preserved without forced reconciliation.
4. **Claims Not Established.** Assertions that must not be made unless further evidence satisfies the applicable authorization requirements.

3 Mathematical Architecture and Version Boundary

The OIP source lineage includes early summary notation and later review-ready formula presentations. OIP v1.0.30 explicitly records that conflicting forms must not be silently reconciled and that a formula must be version-locked before confirmatory implementation. For the consolidation in this edition, the review-ready forms below are reproduced as the v1.0.40 reference notation. This editorial consolidation is not evidence of empirical validity. Any implementation must preserve the version and definitions used.

3.1 Infrastructure Independence

$$\mathrm{II} = 1 - \mathrm{IDS}, \qquad 0 \leq \mathrm{IDS} \leq 1. \tag{1}$$

Here, IDS denotes Infrastructure Dependency Score and II denotes Infrastructure Independence.

3.2 OIP Stability

$$\text{OIP Stability} = \mathrm{GFL} \times \mathrm{II} \times \mathrm{AML} = \mathrm{GFL} \times (1 - \mathrm{IDS}) \times \mathrm{AML}. \tag{2}$$

The source definitions describe GFL as a Grounded Feedback Loop and AML as Adaptive Moral Logic. Formula presence in a specification does not itself establish that either construct has been validly measured in a dataset.

3.3 Living Intelligence Value

$$\mathrm{LIV} = \frac{\mathrm{RI} \times \mathrm{SR} \times \mathrm{AR}}{\mathrm{IDS}}. \tag{3}$$

The source notation associates RI with Resilience Impact, SR with Scalability Range, and AR with Adaptation Rate.

Critical boundary condition

$$\text{IDS} = 0 \quad \Longrightarrow \quad \text{LIV} = \texttt{UNDEFINED}. \tag{4}$$

No epsilon floor, denominator replacement, clipping, or arbitrary lower bound is authorized to make this expression numerically defined. This boundary applies to LIV. It does not make the separate expression $II = 1 - IDS$ undefined at $IDS = 0$.

3.4 Neutral competence score

$$\text{Score} = 0.6\,\text{OE} + 0.3\,\text{CV} + 0.1\,\text{VIS}. \tag{5}$$

where OE is Outcome Efficiency, CV is Contextual Validity, and VIS is Verifiable Impact Stability. The source lineage records earlier alternative notation for this formula. The form above is the reference form selected for this consolidated edition; it is not presented as empirically calibrated or universally valid.

3.5 Survival Intelligence Score

$$\text{SIS} = 0.5\,\text{RC} + 0.3\,\text{AFL} + 0.2\,\text{CAS}. \tag{6}$$

where RC is Resilience Capacity, AFL is Adaptive Feedback Learning, and CAS is Contextual Adaptation Speed, as used in the review-ready source notation. These components require explicit operational definitions before empirical use.

3.6 Stress-response model

$$\text{IS} = 0.1\,\text{SD} + 0.9\,\text{CW}. \tag{7}$$

The source describes this as a situational stress-response weighting model, not a universal constant. It must not be reinterpreted as universally valid weights without new specification and evidence.

3.7 Formula and measurement limits

The equations are specification-level expressions. For constructs whose ranges, normalization, measurement rules, or semantic mappings are not fully established by documentary evidence, operationalization remains unauthorized. A computable expression is not, by itself, a valid measurement or a validation result.

4 Evidence Hierarchy and Authorization Gates

4.1 Seventeen-layer evidence hierarchy

The canonical hierarchy in OIP v1.0.30 is:

1. OIP documented specification
2. Authoritative dataset codebook
3. Questionnaire and study documentation
4. Frozen raw data and provenance

5. Verified variable identity
6. Verified coding and category meaning
7. Routing and eligibility evidence
8. Missingness provenance and meaning
9. Construct semantic alignment
10. Directionality justification
11. Frozen operationalization
12. Frozen final analytical cohort
13. Frozen score calculation
14. Outcome estimability
15. Independent evaluation
16. Uncertainty analysis
17. Evidence-bounded claim

AI-generated suggestions may be used as candidate hypotheses or review prompts. They do not authorize variables, define missingness rules, invent transformations, establish construct validity, or establish empirical validity.

4.2 Six dataset qualification gates

1. **Provenance and integrity.** Record the source identity, provenance, and relevant artifact hashes.
2. **Temporal validity.** Establish the temporal order of predictors, decisions, and outcomes, such as $T_0 \to T_1 \to T_2$.
3. **Construct evidence.** Establish documentary support for GFL, IDS, AML, and any other construct being operationalized.
4. **Outcome independence and estimability.** Define an independent outcome and determine whether the proposed statistic is estimable. For binary ROC-AUC, both positive and negative outcome classes must be present; this condition alone does not establish overall validity.
5. **No synthetic outcome construction.** Do not manufacture outcome labels from predictors, proxies, or the score being evaluated.
6. **Leakage control.** Examine mechanical, temporal, and semantic leakage under a documented design.

4.3 Three scientific gates

1. **Gate A: Score computability.** The required constructs and transformations must be authorized before calculating the score.
2. **Gate B: Outcome evaluation.** The outcome, cohort, and evaluation design must be independent and estimable under the pre-specified rules.
3. **Gate C: Empirical validity claim.** A validity claim requires appropriate evaluation and uncertainty analysis; it cannot be inferred from a successful audit or a computable score.

4.4 Ten-condition authorization conjunction

Downstream scientific evaluation requires all of the following conditions to be satisfied:

$$\begin{aligned}&\text{Source Definitions Frozen} \wedge \text{GFL Approved} \wedge \text{IDS Approved} \wedge \text{AML Approved}\\ &\wedge\ \text{Analytical Cohort Frozen} \wedge \text{Score Frozen} \wedge \text{Outcome Independent}\\ &\wedge\ \text{Outcome Estimable} \wedge \text{Leakage Audit Passed} \wedge \text{Evaluation Design Frozen}.\end{aligned}$$

If any required condition is not established, the corresponding downstream authorization is withheld.

4.5 Missingness rule

$$\texttt{Missing} \neq 0$$

Missing values must not be silently converted to zero. Missingness, routing, and eligibility must be interpreted from authoritative evidence. Duplicates, proxies, transformations, and thresholds must not be changed simply to obtain a preferred result.

5 Empirical Audit Lineage

This section summarizes the versioned audit record. Counts labelled **reported** are reproduced from the available historical records and are not represented as newly recomputed facts unless specifically stated. Unresolved conflicts remain visible.

Table 1: OIP audit lineage and authorization boundaries

Version	Target	Recorded evidence	Recorded boundary
v1.0.28	HR benchmark	Historical report records $N = 54{,}808$, baseline yield 12.66%, OIP yield 20.62%, absolute difference 7.96 percentage points, and reported $p = 3.59 \times 10^{-29}$. Later lineage classifies this record as a synthetic simulation / decision-logic demonstration, not independent empirical validation.	Historical reported metrics only; not current validation evidence.
v1.0.29	BEMP panel	The supplied notebook copy is unexecuted and contains no recorded cell outputs. Earlier summaries report 470 units, 32 positives, zero negatives, and 438 missing or unresolved. These exact runtime counts cannot be reconstructed from the supplied notebook copy.	Fail-closed lineage; ROC-AUC is not estimable if the reported class distribution is correct. Runtime counts remain reported, not independently verified here.
v1.0.31	Malawi IHPS	The v1.0.34 synthesis reports processing 395 Stata files and records `hh_b06_1` as an identifier role. Later inventory summaries contain inconsistent file-count descriptions.	`FAIL_CLOSED_AUDIT_COMPLETE`; row-level longitudinal linkage not established; conflicting inventory counts unresolved.

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Version	Target	Recorded evidence	Recorded boundary
v1.0.32	Uganda UNPS 2019/20	The v1.0.34 synthesis reports 109 Stata files read with zero read errors and 230 candidate outcome variables. Candidate discovery was not promoted to construct or outcome approval.	Constructs and outcomes not approved; score not authorized; empirical evaluation not ready.
v1.0.33	Ethiopia ESS4	75 source files reported: 68 Stata files and 7 PDFs. The record describes 1,446 documentary evidence records and preserves C19 as blocked. The source synthesis records a manifest SHA-256.	Qualification not established; release as audit-only.
v1.0.34	Four-dataset synthesis	Descriptive synthesis of BEMP, Malawi IHPS, Uganda UNPS, and Ethiopia ESS4. It introduces no new construct operationalization, score, outcome, or evaluation.	Evidence-boundary synthesis; not a fifth dataset audit and not a validity claim.
v1.0.35	Archival freeze	A prior summary reports a freeze of 12 upstream artifacts and four conflicts, with a C12 hash. The full manifest has not been independently reconstituted in this edition.	Archival-freeze result is reported; confirm against the corresponding primary manifest before reuse.
v1.0.36	Tanzania NPS Wave 4	Prior summaries report 83 artifacts, comprising 75 Stata files and 8 PDFs, and 27,214 diagnostic rows. Conflicting manifest-hash renderings have appeared in the available lineage.	Fail-closed, audit-only status reported; canonical hash unresolved.
v1.0.37	Nigeria GHS Wave 5	Prior summaries report 121 source files, 108 Stata files, 13 PDFs, and 3,371 unique variable names. Local Version 01 documentation conflicts with the mounted dataset and Basic Information Document Version 2. Conflicting manifest hashes also appear in the lineage.	Fail-closed state retained; identity conflict and canonical hash unresolved.
v1.0.38	Tanzania KHDS 1991–1994	Prior records report 442 files, including 420 physical DTA files, and 2,980 unique variable names with 12,029 occurrences. A count of 339 structural four-wave pairs appears in prior summaries but is not promoted here to a verified final cohort count.	Prior record states `PASS FINAL MASTER AUDIT LOCK`; zero constructs authorized and zero scores calculated. Pair count requires source-level reconciliation before being treated as verified.
v1.0.39	Waymo safety claim audit	Prior audit summaries report CSV3 as 1,804 by 21 and CSV4 as 18,352 by 7, with 601 repeated grains, 1,202 repeated rows, 888 previously mapped values, and 314 benchmark values outside the boundary. The primary audit package was not rerun for this edition.	C08 fail-closed; C09 entry not authorized; execution blocked; scientific authorization false; safety verdict not issued.

5.1 Interpretation of the audit lineage

The lineage documents repeated decisions to withhold downstream authorization where evidence requirements were not satisfied or source conflicts remained unresolved. The audits do not establish universal measurability or unmeasurability, empirical validity, predictive superiority, causal validity, or universal applicability of OIP.

6 Synthetic Software Reference Verification: OIP v0.2.0

OIP v0.2.0 is a synthetic Python reference implementation for selected authorization decision rules. It is distinct from the broader OIP v1.0.x methodological and dataset-audit lineage. It is not a production agent, a new language model, or a complete implementation of every OIP concept.

6.1 Recorded test scopes

- The baseline verification export contains 114 planned test records plus 14 supplemental core-reason records, for 128 exported data rows.
- The `Exhaustive_Matrix` section records 72 passing matrix entries. The reference expectations were encoded in the test artefacts and were not independently audited as a separate oracle.
- Initial release documentation separately reports a 26-test unit-test scope. This is a separate reported scope and must not be added to 128 or described as the same count without a reconciliation of test definitions and runs.
- The implementation records explicit decision states including `APPROVED`, `PENDING_EVIDENCE`, `BLOCKED`, and `INVALID_INPUT`.

6.2 Export artifact integrity

The SHA-256 values below were recomputed from the supplied JSON and CSV artifacts for this edition and matched the recorded values.

JSON SHA-256

`a1f387bcf9a855ea9a9cc233f9697661274d871d45ce85d0347e7f564c6b17bd`

CSV SHA-256

`b4637bda0989e66fe3bf20428340f29a7f0a4269ad31a063ca7a1f2b8c576b84`

6.3 What these tests establish and do not establish

The tests establish the recorded behavior of the specified synthetic inputs under the tested implementation and test expectations. They do not establish scientific validity, causal validity, real-world agent safety, security certification, production readiness, robustness under arbitrary adversarial conditions, or compliance with any later OIP version. The SHA-256 comparison establishes the identity of the compared files, not the correctness of the protocol.

7 Integration Governance and Licensing Boundary

This section states proposed integration controls for any future evaluation or commercial licence. These provisions are not, by themselves, an executed contract and do not grant a company permission to use OIP. Binding terms require a separate agreement that identifies the licensed materials, permitted use, term, fees, responsibilities, and governing law.

7.1 Proposed integration controls

1. **Fail-closed execution.** If a required evidence gate fails, the integrated system must prevent the affected downstream operation and preserve an explicit state such as `PENDING_EVIDENCE` or `BLOCKED`.
2. **Missingness integrity.** Missing values must not be silently converted to zero. Proxy substitution, unsupported imputation, and post-hoc adjustment of thresholds or weights must not be used to manufacture a preferred finding.
3. **Auditability.** Implementations should record source identifiers, version, gate results, state transitions, input/output references, and relevant integrity hashes. Logs should be reproducible within the documented environment; a hash alone does not make storage immutable.
4. **AI boundary.** AI-generated variable mappings, interpretations, and methodological suggestions remain candidates until approved by documentary evidence and the applicable review process.
5. **Claim discipline.** No adopter should represent OIP as empirically validated, causally validated, universally valid, superior to alternatives, commercially validated, production-ready, or hardware-authorized unless separate evidence supports the specific claim.
6. **Independent safety controls.** OIP should not replace authentication, access control, sandboxing, monitoring, human approvals, security testing, or other controls appropriate to the deployment.

7.2 Separate licence required

A party seeking to use or integrate OIP must obtain written authorization under a separate licence or agreement from the relevant rights holder. This technical reference does not determine the legal status of every formula, idea, source file, third-party component, or historical document in the OIP record. Rights, warranties, limitations of liability, indemnities, and dispute terms must be addressed in the separate agreement after appropriate legal review.

8 Limitations, Corrections, and Open Evidence

The following unresolved items must remain visible until the relevant primary records are obtained and reconciled:

1. Reconstruct the exact runtime outcomes for the supplied BEMP notebook from an executed notebook or archived result package.
2. Reconcile Malawi file inventories and state exactly which count refers to physical DTA files, directory entries, manifest rows, or processed files.
3. Verify the canonical manifest hash for Tanzania NPS Wave 4 from the corresponding original manifest and its byte-level file.

4. Verify the canonical Nigeria GHS Wave 5 manifest and retain the documented Version 01 versus Version 2 identity conflict unless primary evidence resolves it.
5. Re-establish the reported KHDS structural-pair count from source files and preserve the rule that structural overlap does not authorize an analytical cohort.
6. Obtain the complete Waymo primary audit package before describing its reported counts as independently reproduced in this edition.
7. Reconcile the 26-test release scope with the 114 planned records, 14 supplemental records, and 72-case matrix by mapping record identifiers to distinct test definitions and execution runs.
8. Complete the OIP v0.2.0 repository and archive licence review before offering commercial terms to an adopter.

These items are deliberately not filled by inference. A later correction should identify the exact source, the affected claim, the previous value, the replacement value, and the reason for the change.

9 Final Research Position

OIP is a research-stage methodology whose principal objective is to maintain an explicit evidence boundary between source material, construct measurement, analytical authorization, evaluation, and scientific claims without manufacturing artificial certainty.

The available record supports the existence of a documented methodology, a series of fail-closed dataset audit records, and a synthetic software reference implementation with reproducible export-file hashes. It does not establish that OIP has been independently validated as a general-purpose AI safety control or that it is ready for production or commercial deployment.

9.1 Explicit non-claims

The current evidence does not establish that OIP is:

- empirically validated;
- predictively proven;
- causally validated;
- universally valid;
- superior to alternative methods;
- commercially validated;
- production-ready; or
- authorized for hardware or safety-critical deployment.

9.2 Closing principle

Evidence must authorize measurement.

Measurement must authorize scoring.

Scoring must precede evaluation.

Evaluation must precede claims.

Document control

Version: OIP v1.0.40

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Author: Sudharsan Das

Location: Tripura, India

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This document records the evidence boundary as of the release date. Corrections require an explicit versioned record.

A Source and Artifact Register

The following source classes inform this consolidation. The register describes the role of each source and does not imply that every historic raw dataset or notebook was re-executed for this release.

1. **OIP v1.0.26.** Foundational conceptual specification and initial mathematical architecture. Historical wording and formula variants are retained as version history rather than silently treated as fully validated results.
2. **OIP v1.0.27.** Mathematical provenance and completion register, including unresolved operationalization items.
3. **OIP v1.0.28.** Historical HR benchmark report. Later lineage describes it as a synthetic simulation / decision-logic demonstration rather than independent empirical validation.
4. **OIP v1.0.29.** BEMP audit artefacts. The supplied notebook copy described in v1.0.34 has no executed outputs, so exact runtime counts are not reconstructed from that copy.
5. **OIP v1.0.30.** Normative construct measurement and dataset qualification protocol. It defines the evidence hierarchy, qualification gates, state machine, and claim discipline.
6. **OIP v1.0.31–v1.0.39.** Dataset audit reports, synthesis records, archival freeze artefacts, and the Waymo safety-claim audit, subject to the evidence qualifications stated in Section 5.
7. **OIP v1.0.34.** Four-Dataset Audit Synthesis and Evidence Boundary Report, September 2026. This source explicitly states that the synthesis introduces no new dataset, construct operationalization, score, outcome, or validity claim.
8. **OIP v0.2.0 JSON and CSV exports.** The supplied exports whose SHA-256 hashes were recomputed for this edition. Their integrity hashes do not establish broader scientific or security validity.

A.1 Public research records

- Zenodo record: `https://zenodo.org/records/23635920`
- GitHub repository: `https://github.com/sudharsandas27-ux/Sudharsan-Das`